Hints: Functional sequences, series and spaces – 4. Spaces of functions

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Completeness: a Cauchy sequence in the uniform norm converges uniformly (Cauchy criterion), and the limit is continuous. Separability: polynomials with rational coefficients (Weierstrass), or piecewise linear functions with rational nodes and values (uniform continuity), form a countable dense set.

Hint 2. Non-compactness of the ball: tents φ_n of height 1 supported on disjoint intervals $\left[\frac{1}{n+1}, \frac{1}{n}\right]$ satisfy $\|\varphi_n - \varphi_m\| = 1$ for $n \neq m$, so no subsequence is Cauchy.

Hint 3. Riesz's lemma: for a closed proper subspace Y and $\theta < 1$, take $x_0 \notin Y$, $d = \operatorname{dist}(x_0, Y) > 0$, and $y_0 \in Y$ with $\|x_0 - y_0\| \leqslant d/\theta$; then $x = \frac{x_0 - y_0}{\|x_0 - y_0\|}$ has $\operatorname{dist}(x, Y) \geqslant \theta$. In infinite dimensions apply it repeatedly to the (closed, because finite-dimensional) spans of $x_1, \ldots, x_{n-1}$, getting unit vectors with pairwise distances at least $\frac{1}{2}$.

Problem 2

Hint 1. Banach: $d(x_{n+1}, x_n) \leqslant q^n d(x_1, x_0)$, so (x_n) is Cauchy, with geometric bound $d(x_n, x) \leqslant \frac{q^n}{1-q} d(x_1, x_0)$; the limit is fixed by continuity, and two fixed points satisfy $d(x, y) \leqslant q\, d(x, y)$.

Hint 2. Picard: on a box $|x - x_0| \leqslant a$, $|y - y_0| \leqslant b$ let $|f| \leqslant M$ and L be the Lipschitz constant. On the closed subset $\{y \in C[x_0 - h, x_0 + h] : |y - y_0| \leqslant b\}$ the map $Ty(x) = y_0 + \int_{x_0}^{x} f(t, y(t))\, dt$ maps the set into itself if $hM \leqslant b$, and is a contraction if $hL < 1$. Fixed points of T are exactly the solutions (fundamental theorem of calculus).

Hint 3. Without Lipschitz: $y' = 3y^{2/3}$, $y(0) = 0$ has the solutions $y \equiv 0$ and $y = x^3$ (and many more glued ones).

Problem 3

Hint 1. Use the Faber–Schauder system: $\varphi_0 = 1$, $\varphi_1 = x$, and then, for $k \geqslant 0$ and $0 \leqslant j < 2^k$, the tent of height 1 supported on $\left[\frac{j}{2^k}, \frac{j+1}{2^k}\right]$ with peak at the midpoint, ordered level by level.

Hint 2. Choose the coefficients so that after finishing level k the partial sum is the piecewise linear interpolant $L_k f$ of f at the dyadic points $\frac{i}{2^{k+1}}$: the coefficient of a new tent is f(midpoint) minus the average of f at the two endpoints. Uniform continuity gives $\|f - L_k f\| \leqslant 2\, \omega(f, 2^{-k-1}) \to 0$.

Hint 3. Partial sums inside a level are again piecewise linear interpolants of f at a set of nodes of mesh at most 2^{-k}, so the same estimate holds for every partial sum, not just the block sums. Uniqueness of the coefficients: evaluating a uniformly convergent expansion at the dyadic points recovers them one by one.

Problem 4

Hint 1. Non-completeness: continuous f_n equal to 0 on $[0, \frac{1}{2}]$, to 1 on $[\frac{1}{2} + \frac{1}{n}, 1]$ and linear in between are ρ-Cauchy. If $\rho(f_n, g) \to 0$ with g continuous, then $\int_0^1 |g - \mathbf{1}_{(1/2,1]}| = 0$, which forces $g = 0$ on $[0, \frac{1}{2})$ and $g = 1$ on $(\frac{1}{2}, 1]$, impossible.

Hint 2. Build the completion: W is the set of ρ-Cauchy sequences modulo $(f_n) \sim (g_n) \iff \rho(f_n, g_n) \to 0$, with $\tilde{\rho}([f_n], [g_n]) = \lim \rho(f_n, g_n)$ (the limit exists because $|\rho(f_n, g_n) - \rho(f_m, g_m)| \leqslant \rho(f_n, f_m) + \rho(g_n, g_m)$). Addition and scalar multiplication are termwise and well defined because ρ comes from a norm.

Hint 3. The embedding is $f \mapsto [(f, f, \dots)]$; it is an isometry, and its image is dense because $[(f_n)]$ is the $\tilde{\rho}$-limit of the constant sequences $[(f_k, f_k, \dots)]$ as $k \to \infty$. For completeness of W, approximate a $\tilde{\rho}$-Cauchy sequence w_k by elements f_k of the image with $\tilde{\rho}(w_k, f_k) < \frac{1}{k}$; then (f_k) is ρ-Cauchy and its class is the limit. (Concretely W is $L^1[0, 1]$.)

Problem 5

Hint 1. Let $c = T(1)$. Since T commutes with differentiation, $c' = T(1') = T(0) = 0$, so c is a constant function.

Hint 2. Fix a point a. By Hadamard's lemma every $f \in C^\infty(\mathbb{R})$ can be written $f(x) = f(a) + (x - a)g(x)$ with $g(x) = \int_0^1 f'(a + t(x - a))\, dt$ smooth. Apply T, using linearity and the commutation with multiplication by x (hence by $x - a$): $Tf = cf(a) + (x - a)\, Tg$. Evaluating at $x = a$ gives $(Tf)(a) = cf(a)$.